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Decentralized framework for product security using blockchain and smart contract-based QR verification

Shivanna K¹, Jayantkumar A. Rathod², Janardhana D R^{3*}, Vidyasagar K B⁴, Vasudev S. Shahapur⁵ and Chaitra Nayak J³

*Correspondence:

Janardhana D R
jdr5414@gmail.com

¹Department of Computer Science and Engineering, Dayananda Sagar Academy of Technology and Management, Bengaluru, India

²Department of Computer Science and Engineering, ATME College of Engineering, Mysuru, India

³Department of Computer Science and Engineering, Nitte Meenakshi Institute of Technology (NMIT), NITTE (Deemed to be University), Bengaluru, India

⁴Department of Computer Science and Engineering, BGS Institute of Technology, Adichunchanagiri University, Mandya, India

⁵Department of Computer Science (IoT, Cybersecurity and Blockchain), Alva's Institute of Engineering and Technology, Moodubidri, India

Abstract

Counterfeiting of goods has become a significant problem across various sectors, ranging from medical to luxury products. This paper introduces a blockchain-based product authentication and security system to combat fraudulent goods and promote supply chain visibility. Leveraging blockchain's decentralization, the system enables tamper-proof tracking of product information and warranty authenticity, from the production level to the end consumer. The proposed system employs technologies such as smart contracts, QR code-based verification, and cryptographic hashing to ensure data integrity. In addition, the paper discusses the scalability limitations of applying blockchain to large-scale supply systems and addresses issues related to privacy and regulations. We explain how to overcome these challenges without compromising the system's security and effectiveness. The model is designed to be scalable, flexible, and cost-effective for industries with complex supply chains, including food safety, luxury goods, and pharmaceuticals. Case studies demonstrate the potential of this anti-fraud infrastructure to enhance product traceability and foster consumer trust. The paper also explores emerging areas of blockchain research in supply chain management and possible enhancements for broader adoption across industries.

Keywords Blockchain, Product authentication, Smart contract, QR code verification, Counterfeit products, Supply chain

1 Introduction

The proliferation of fake products is a global issue impacting consumers, businesses and economies everywhere. Counterfeit items, such as bags, clothing, electronics and cosmetics, mislead buyers and can pose serious health and safety risks. These fraudulent activities result in financial losses and tarnish the reputation of legitimate companies, leading to customer dissatisfaction and diminished trust. According to The State of Counterfeiting in India 2021 by Authentication Solution Providers' Association (ASPA), the Indian economy suffers a loss of approximately INR 1 trillion annually due to fraudulent supply chain items, with cases increasing by about 20% from 2018 to



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2020. Traditional methods to combat unauthorized replication of goods, like bar codes and holograms, often fall short in providing the necessary security and transparency to ensure product authenticity. Therefore, there is an urgent need for innovative solutions to effectively tackle this issue. Blockchain technology presents a promising solution for identifying and preventing counterfeit goods. It is a decentralized digital ledger that is secure and provides maximum transparency for transactions. By utilizing blockchain, companies can create a complete digital record of a product's origin and track it throughout the supply chain. Each transaction related to the product is securely recorded on the blockchain, forming an unalterable and transparent history. This digital trail enables companies to verify the authenticity of products at any point in the supply chain, boosting consumer confidence in their purchases. Adopting blockchain technology to combat fake products enhances network transparency, controls cost and manages supplier relationships. It enables end customers, suppliers and vendors to securely track a product's supply chain with minimal risk of encountering counterfeit items. This paper intends to present a blockchain system designed to prevent brand illegitimate goods, featuring a robust model to ensure that products are genuine and authentic. This framework combines the strengths of blockchain decentralization, smart contracts, and QR code-based verification in one unified approach to product security. Unlike prior solutions, it actively maintains tamper-proof records of every product's lifecycle, from manufacturing to sale, using immutable blockchain storage and cryptographically secure event updates. The system assigns specific roles and privileges—administrator, manufacturer, supplier, retailer, and consumer—with each participant using smart contracts to record and verify transactions. Consumers can independently authenticate products by scanning QR codes linked directly to the blockchain, ensuring both transparency and real-time traceability. This layered structure and integration of decentralized verification and smart contracts rise above the static identification methods commonly used, offering dynamic protection and enhanced supply chain visibility.

With fraudulent goods increasingly threatening industries and consumer safety, this solution addresses critical gaps in security and trust. By enabling live, tamper-resistant tracking and instant validation at every supply chain point, the risk of fraudulent products entering the marketplace is drastically reduced. Full transparency and independent authentication empower all stakeholders, especially consumers, while supporting regulatory compliance for businesses. Its scalable, flexible design means it can be adapted for industries with complex supply chains—such as pharmaceuticals, luxury goods, and food safety—delivering both reliability and cost-effectiveness. Ultimately, the framework fosters stronger brand integrity and safer, more trustworthy purchasing experiences.

In this research work, Sect. 2 presents the related work and summarizes existing blockchain-based anti-counterfeiting solutions. Section 3 details the proposed methodology and system architecture. Section 4 reports the experimental results and evaluation findings. Section 5 outlines the key challenges encountered and highlights potential future research directions. Finally, Sect. 6 provides the conclusion of the work.

2 Related works

By analyzing current methods, identifying their limitations and highlighting their strengths, this section establishes a foundation for understanding the research landscape. In the literature review, we examine current methods for verifying products

using blockchain, highlighting any shortcomings and demonstrating how this work will address solving them. In [1] authors discuss how to detect fake products with blockchain technology. It suggests a solution based on blockchain to guarantee product authenticity and traceability. This involves using unique QR codes connected to a blockchain network, allowing consumers to check products by scanning the codes and in [2] paper explores how blockchain technology can identify counterfeit products in the supply chain. It highlights the importance of transparency and authenticity by using unchangeable transaction records and "proof of possession" for items. In The paper [3] investigates how blockchain can be used to detect counterfeits, addressing the limitations of traditional methods such as RFID tags, AI and QR codes. Blockchain provides a secure, clear and decentralized way to track products. In [4] authors proposed a study combines blockchain technology, QR codes and Firebase to improve supply chain management and verify product authenticity and the applications of financial and health risks addresses the of counterfeit products and suggests using blockchain, AI and machine learning for better counterfeit detection in [5]. In [6, 7] discussed about combination of blockchain with QR codes for better supply chain management, providing traceability and security, mainly for sensitive products like food and medicine. In [8, 9] discussed about testing Solidity smart contracts in open-source projects, highlighting the importance of testing for blockchain-based systems and NFT-based solution for counterfeit detection, especially in industries like pharmaceuticals, electronics and luxury goods.

In [10] authors investigate the development of a secure tracking and tracing system by integrating blockchain and QR code technologies to enhance security and reduce computational time. In [11] Proposed a blockchain-based framework for multi-tier textile and clothing supply chains, offering transparent, tamper-proof records and improved operational efficiency by eliminating intermediaries. In [12] emphasized blockchain's role in promoting sustainability through resource optimization, waste reduction, and ethical sourcing. In [13] they highlighted blockchain's ability to enhance traceability, security, and efficiency while addressing challenges like scalability and regulation. Similarly, in [14] demonstrated how blockchain improves transparency, consumer trust, and regulatory compliance across sectors such as food safety and pharmaceuticals. The role of multi-stakeholder initiatives (MSIs) discussed in [15] promoting sustainability in supply chains. The study shows how MSIs foster collaboration among corporations, NGOs and governments, providing guidelines and metrics to improve environmental, social and economic outcomes, with industry-specific and regional factors influencing their effectiveness and blockchain-based solution to combat counterfeit products, addressing consumer risks and manufacturer challenges like revenue loss and brand damage in [16, 17]. Blockchain ensures data immutability and transparency, enabling secure product verification through QR codes linked to the blockchain. A camera scanner verifies these QR codes, instantly notifying consumers about product authenticity. This approach reduces dependence on retailers, empowering consumers to independently validate product genuineness and demonstrates blockchain's effectiveness in curbing counterfeits and highlights the supply chain transparency challenges such as service duplication, poor coordination and lack of standardization. Unauthorized replication of goods remains problematic, with traditional technologies like RFID, AI and QR codes facing issues like duplication and high computational demands.

To address these, the authors suggest linking product serial numbers with customer IDs in the seller's system through blockchain. This decentralized, tamper-proof system ensures data immutability, improves traceability and prevents unauthorized modifications, hence offering a reliable solution for counterfeit detection [18]. In [19–21] address the commonness of counterfeit products in online stores, where nearly 20% of goods are reported as fake. These counterfeits harm consumers, damage brand reputations and lead to financial losses for businesses. The authors propose a blockchain-based system for product authentication, showing its transparency and immutability to store and verify product details via QR codes. This decentralized approach reduces intermediaries; lowers quality assurance costs and helps consumers verify authenticity by themselves. The study highlights blockchain's power to transform counterfeit detection and improve trust between consumers and manufacturers.

In [22, 23] authors proposed a blockchain-based solution for the Agri-Food supply chain, enhancing traceability, transparency and trust. Using Ethereum smart contracts and IPFS, the system ensures secure, immutable data storage for efficient product tracking. It uses a reputation system for independent transactions to address fraud and accountability issues. Security analyses show how well the system can defend in case of attacks, proving it has better alternatives to currently existing solutions. In [24] proposes Lelantos, a blockchain-based system for anonymous e-commerce delivery, addressing privacy concerns. Using Ethereum smart contracts and decentralization, Lelantos ensures customer anonymity, fair exchanges and unlink ability between customers and merchants. The system combines on-chain smart contracts with off-chain components, that follows a process of passing the package through multiple companies so that there is no link. Customers can pick up packages from any route location, ensuring privacy and security in case of sensitive purchases.

A blockchain-based Product Ownership Management System (POMS) addressed in [25–27] to overcome spurious products in the post-supply chain, addressing the limitations of RFID tags, that can be cloned. Their system uses blockchain to enable proof of product ownership, allowing customers to verify authenticity even if the RFID tag is genuine. Ethereum is used in the system, and it ensures that sellers must prove ownership, providing a cost-effective solution (under \$1 per transfer) for securing the product and preventing production of fake items. In [28] evaluate the performance of Ethereum private and testnet networks using Hyperledger Caliper, addressing challenges in performance testing. They highlight the limitations of using Ethereum's mainnet due to real Ethers and differences between private networks and the main network. Comparing the Ethereum private network with the Ropsten testnet, the study finds that the private network outperforms the unstable Ropsten testnet, offering valuable insights for improving the performance of decentralized applications (DApps) on Ethereum and in [29] explores the use of Ethereum smart contracts in managing food supply chains, addressing challenges such as transparency, security and traceability. Blockchain provides a decentralized, immutable ledger, while smart contracts automate transactions, enhancing trust and eliminating intermediaries. The study highlights the use of Truffle and Ganache frameworks for developing, testing and deploying smart contracts, ensuring secure and efficient operations. These technologies contribute to solving issues like food contamination and fraud in the supply chain.

In the paper [30] explores Ethereum, a blockchain system designed to support decentralized, transaction-based state machines. It introduces key concepts like cryptographically secure transactions, a generalized transaction model and Ethereum's proof-of-work consensus mechanism. The paper discusses Ethereum's potential to enable decentralized applications and its native currency, Ether, which incentivizes computation. Wood emphasizes Ethereum's ability to provide a trustless, transparent alternative to traditional centralized systems. In [31] proposed a model to predict the performance and storage of Ethereum contracts based on transaction volume. By analyzing Ethereum's "World State," the model establishes a relationship between transaction volume, execution time and storage requirements. This approach helps companies forecast system performance and storage needs, addressing challenges of performance degradation and large storage occupation as transaction volumes increase in Ethereum-based consortium blockchains.

In the paper [32] presents a comprehensive analysis of Ethereum smart contracts, focusing on both the Solidity source code and the associated opcodes. By studying tens of thousands of verified smart contracts, the authors aim to understand how programmers utilize the Ethereum language instructions. Their findings show that most contracts have a simple, linear structure with minimal use of control loops or inter-contract calls. Additionally, the study identifies key opcodes used in discarding those that are rarely deployed. This research lays the foundation for development of domain-specific languages (DSLs) to assist in the creation and verification of decentralized applications (DApps). In [33–35] examine the shift from centralized to decentralized digital identity, focusing on Self-Sovereign Identity (SSI) powered by blockchain. They discuss the limitations of traditional identity systems, such as lack of user control and security risks and how blockchain addresses these issues with its decentralization, tamper-proof and traceable nature. The paper compares various SSI implementation approaches and identifies challenges, offering insights into future developments in blockchain-based digital identity systems, driven by the growing need for privacy and security in the digital age.

The Historical Weighted Difficulty (HWD) protocol to mitigate 51% attacks on Proof-of-Work (PoW) blockchains proposed in [36]. The protocol calculates the total difficulty of a blockchain branch by considering the distribution of miner addresses over a history window, with greater weight assigned to miners who have contributed to the blockchain over time. This makes it harder for new miners or attackers with temporary hash power to create fraudulent branches, as their efforts are less influential compared to established miners. The analysis demonstrates that this method can increase the cost and time required for a successful 51% attack by up to two orders of magnitude, thereby significantly strengthening the blockchain's security. In [37] propose the Lib-Pri system, a lightweight blockchain-based privacy protection scheme for smart surveillance. It integrates blockchain, smart contracts and object detection to preserve privacy in video surveillance. Smart cameras at the edge apply reversible blurring based on privacy policies, while the blockchain ensures authenticated access. Off-chain storage is used for important video data, and the system can identify fugitives by comparing facial features. This decentralized approach enhances privacy while maintaining surveillance security.

The performance evaluation of the Hyper ledger Fabric blockchain framework done in [38–40], focusing on throughput, latency and scalability. The study explores how network workload, including transaction rates, transaction numbers and types, impacts the

performance of the platform. Using Hyperledger Caliper as the benchmarking tool, the authors test various scenarios involving transaction rates, transaction numbers and simultaneous transactions. Results show that Hyperledger Fabric can support up to 100,000 participants and handle transaction rates of up to 200 transactions per second without significant latency. The study highlights the relationship between transaction types and system performance, providing insights into optimizing blockchain network configurations for efficiency. In [41] proposed a private blockchain-based system for securing Electronic Medical Records (EMRs). The study emphasizes enhancing healthcare data management, ensuring secure data transfer and maintaining patient confidentiality. The blockchain system establishes seamless integration between health departments and hospitals, streamlining access and improving safety.

The reviewed literature collectively explores the application of blockchain technology for product verification, counterfeit detection, and supply chain transparency across various domains, including food, pharmaceuticals, textiles, and e-commerce. Most studies emphasize blockchain's strengths—immutability, transparency, and decentralization—in addressing the limitations of traditional authentication techniques such as RFID, AI, and QR codes, which suffer from duplication and high computational overhead. Several works integrate blockchain with smart contracts, QR codes, and support technologies like Firebase, IPFS, and AI to enhance traceability and trust within supply chains. Other research focuses on system performance, such as improving efficiency in Ethereum and Hyperledger frameworks or ensuring scalability and low latency. While some studies extend blockchain's use to areas like digital identity, privacy preservation, and sustainability, few offer a unified comparison of methodologies or evaluate trade-offs in terms of cost, computational performance, or implementation complexity. The existing body of work, though diverse, reveals a gap in comprehensive analysis and synthesis—particularly in categorizing approaches by their objectives (e.g., authentication, privacy, performance optimization) and identifying unresolved issues such as interoperability, standardization, and scalability. This gap underscores the need for a structured comparative study that bridges theoretical frameworks with practical implementation insights to advance blockchain-based counterfeit detection.

3 Proposed methodology

TruSecure is a blockchain-based solution designed to combat sale of falsified products by ensuring product authenticity and transparency throughout the supply chain. The system leverages the immutable and decentralized nature of blockchain technology to provide a tamper-proof method for verifying product details. It involves five key roles: Administrator, Manufacturer, Supplier, Retailer and Consumer, each with distinct privileges and responsibilities.

- **Administrator:** Manages user accounts, ensuring that only verified entities participate in the system.
- **Manufacturer:** Logs in with credentials provided by the administrator to connect a MetaMask wallet and register products by entering details like serial number and description. A transaction fee is paid and a QR code is generated for the product.
- **Supplier:** Logs in and accesses options like the manufacturer. To update product details (e.g., location or status), the supplier scans the QR code, confirms details and pays a transaction fee to record changes on the blockchain.

- **Retailer:** Functions like the supplier but have an additional option to mark products as sold. Retailers update product details by scanning the QR code, confirming changes and paying a transaction fee.
- **Consumer:** Accesses the system by scanning the QR code on the product. If authentic, the product's status is displayed. If the product is not authentic, then a warning is displayed. Consumers can view the supply chain details by connecting their wallet.

The proposed system leverages blockchain technology and QR codes to ensure product authenticity throughout the supply chain. The methodology involves the following key steps:

- **QR code generation and registration:** The manufacturer generates a unique QR code for each product, embedding essential details such as product ID, manufacturing date and location. This QR code is then registered on the blockchain, ensuring tamper-proof storage of product records.
- **Supplier verification:** The supplier scans the QR code to access and update product details on the blockchain. This step ensures transparency and traceability across the supply chain.
- **Blockchain storage:** All updates, including location changes and transaction history, are immutably stored on the blockchain, forming a verifiable chain of custody.
- **Customer verification:** The customer scans the QR code using a web interface. The system fetches the product history from the blockchain, verifying details such as manufacturing origin, supplier updates and location history.

The Fig. 1 illustrates a blockchain-based product authentication workflow. The admin sets up accounts for manufacturers, suppliers, and retailers, who each log in, connect their wallets, add or update product information, and generate QR codes for products. Consumers scan these QR codes to verify authenticity; if verified, they can connect their

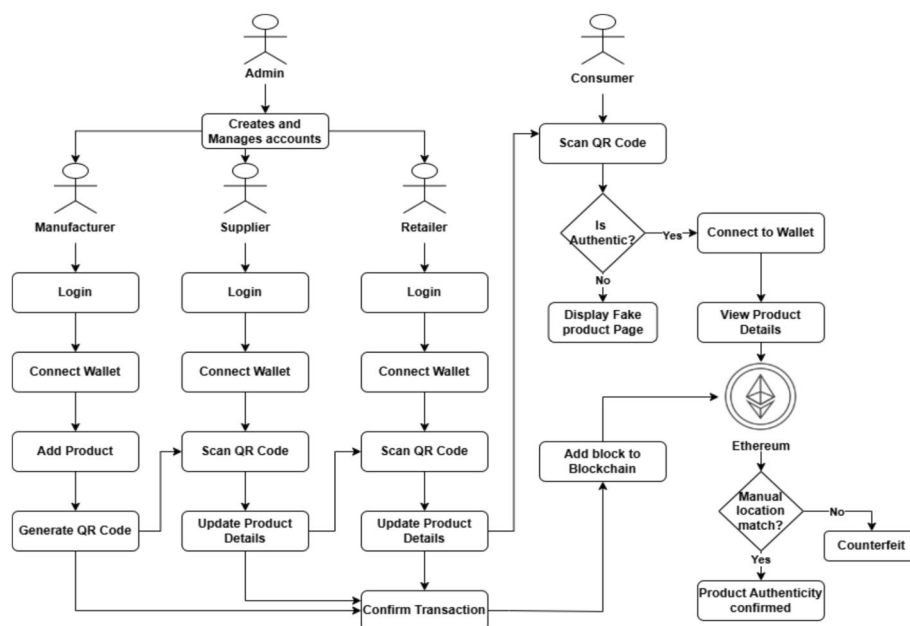


Fig. 1 Architecture and operational workflow of the proposed blockchain-based product authentication system

wallets and view product details on the blockchain. If the product fails authentication or location matching, the workflow flags it as counterfeit; otherwise, authenticity is confirmed and recorded on Ethereum blockchain.

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The Fig 2 shows how a blockchain-based system uses location-based verification to ensure product authenticity and prevent fake products. The process starts at the

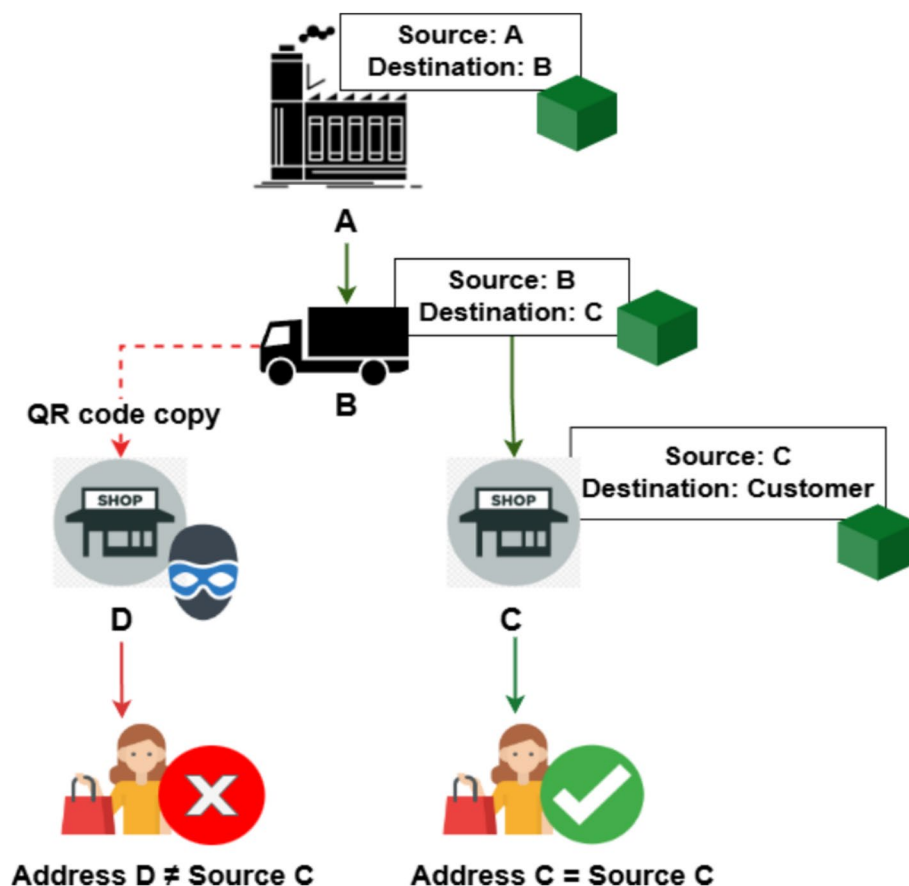


Fig. 2 Workflow illustrates the customer's manual location verification process within the proposed system

manufacturer (Location A), where the product is registered on the blockchain and a unique QR code is generated. The product then moves to the supplier (Location B) and the retailer (Location C), with each updating the product's location on the blockchain. When a customer purchases a product, they scan their QR code. The customer will check if the scanned location (C) matches the location of purchase. If it matches ($C = C$), the product is authentic. However, if a counterfeit product is sold by a fake retailer (e.g., Location D) using a copied QR code, the mismatch in locations ($C \neq D$) will flag the product as fake. This seamless integration of blockchain ensures data transparency and immutability. The use of QR codes allows consumers to verify product authenticity instantly, while supply chain entities ensure accurate product tracking through each stage.

3.1 Proposed architecture

Th Fig. 3 illustrates how address matching thwarts counterfeiting by comparing the source and destination throughout product movement; if these addresses do not align with blockchain records, the product is deemed counterfeit, but a match confirms genuine authenticity.

3.2 Blockchain based traceability algorithm

- i. Hash generation

$$H = \text{Hash}(P) \quad (1)$$

where:

P = Product details (e.g., name, manufacturer, serial number).

H = Unique hash for the product, generated using a secure cryptographic hash function (e.g., SHA-256).

The product details P are passed through a cryptographic hash function to create a unique identifier H , which serves as a digital fingerprint for the product.

Example of Hash generation:

Given product details:

$$P = \{Name : "Laptop", Serial : "12345"\}$$

Hash computation:

$$H = \text{SHA} - 256("Laptop12345")$$

Output:

$$H = "a9c0..."(Aunique256 - bithash)$$

- ii. Blockchain transaction creation

$$T_i = (P, H, T_s) \quad (2)$$

where:

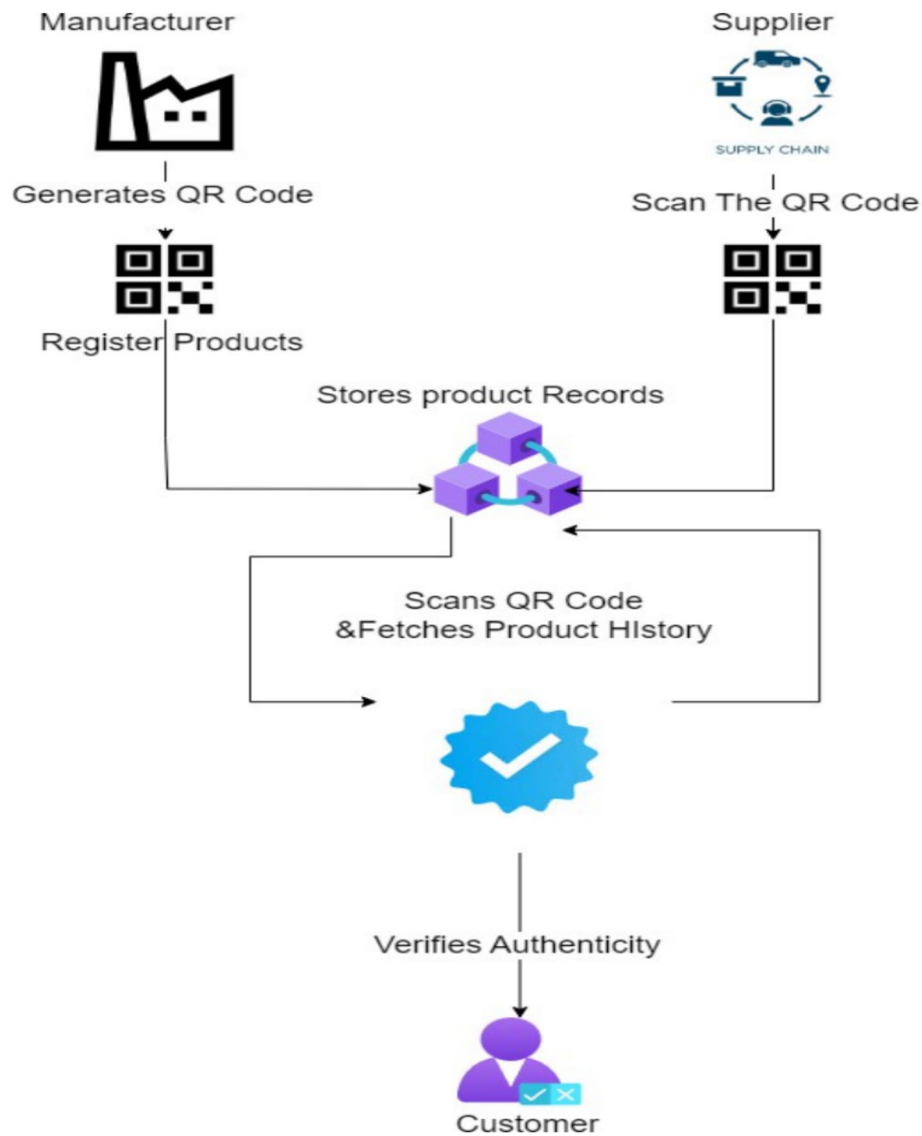


Fig. 3 Architectural representation of the proposed blockchain-based product authentication system

T_i = The i -th blockchain transaction.

P = Product details.

H = Generated product hash.

T_s = Timestamp of the transaction.

A blockchain transaction stores the product details, the generated hash, and a time-stamp, establishing an immutable record of the product's state at a specific point in the supply chain.

iii. Recalculations and comparison of hashes

$$H' = \text{Hash}(P + E) \quad (3)$$

$$\text{Verify: } H' = H$$

where:

H' = Recalculated hash with possible updates.

P = Original product details.

E = Supply chain event details (e.g., location, movement, time).

During supply chain events, a new hash H' is computed using the current product details and recent event information. If $H' = H$, the product remains unaltered and authentic. If $H' \neq H$, integrity has been breached.

iv. Traceability validation

$$\text{Authenticity} = \begin{cases} \text{True}, & \text{if } H' = H \\ \text{False}, & \text{otherwise} \end{cases} \quad (4)$$

The product is validated as authentic if and only if the recalculated hash matches the blockchain-stored hash.

3.3 QR code based validation algorithm

i. QR code data extraction

Algorithm:

$$D = \text{Decode}(\text{QR Code}) \quad (5)$$

Where:

D is the decoded raw data obtained from scanning the QR code.

Extract information:

$$P_{QR} = \text{Parse}(D) \quad (6)$$

where:

P_{QR} is the product information extracted by parsing the decoded data D .

ii. Comparison of QR code data

$$\text{MatchStatus} = \begin{cases} \text{True}, & \text{if } P_{QR} = P_{\text{Blockchain}} \\ \text{False}, & \text{otherwise} \end{cases} \quad (7)$$

Compare product details retrieved from the QR code P_{QR} with the product details P stored on the blockchain to determine consistency and authenticity.

3.4 Integrated system workflow

Step 1: Generate a unique hash H for the product:

$$H = \text{Hash}(P) \quad (8)$$

Step 2: Store the hash H and product details P as a blockchain transaction:

$$T = (P, H, T_s) \quad (9)$$

Step 3: During product movement, update the blockchain with supply chain events E:

$$T \leftarrow T + E \quad (10)$$

Step 4: Allow the end user to scan QR code, retrieve P_{QR} , and validate against blockchain data:

Recalculate hash:

$$H' = \text{Hash}(P_{QR} + E) \quad (11)$$

Compare hashes H' and H :

If $H' = H$, mark the product as authentic.

Step 5: Return the verification status:

$$\text{Status} = \begin{cases} \text{Authentic,} & \text{if } H' = H \\ \text{Compromised} & \text{otherwise} \end{cases} \quad (12)$$

How it works:

Blockchain layer: Ensures immutability and traceability by recording product events and verifying their integrity using hash-based cryptography.

QR Code layer: Provides a quick and accessible way for users to validate product authenticity at the consumer level.

Integrated verification: Combines hash values for verification from the blockchain with data consistency from the QR code to ensure a robust anti-counterfeiting system.

3.5 Technology and tools used

This Fig 4 describes the integration and user flow for a decentralized web application built with React.js and ethers.js, which is hosted on a server. Users access the interface and use MetaMask to connect their wallets; this connection enables them to interact securely with the Ethereum blockchain for executing transactions or verifying information.

A. Technology and Tools used:

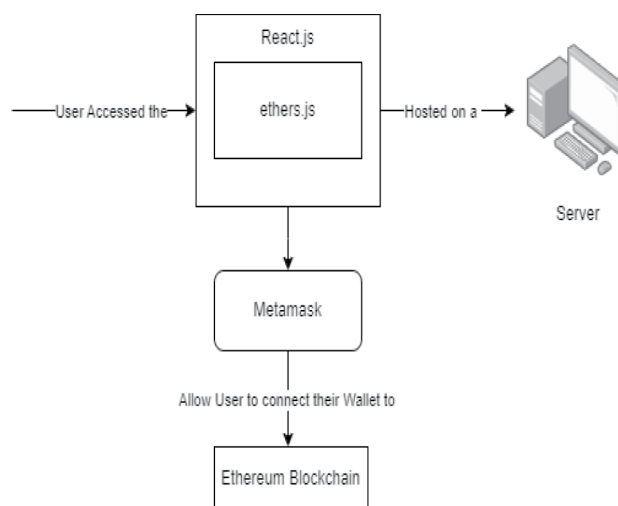


Fig. 4 Data Flow Diagram illustrates the Ethereum wallet connection process within the proposed web application

- *Solidity (Smart contract development)*: Solidity is used to write the business logic behind the blockchain transactions. Smart contracts are deployed on the Ethereum Testnet (Sepolia) and they handle key functions like registering products, updating product history and retrieving data.
- *Hardhat (Development environment)*: In this project, the Hardhat environment plays a crucial role in building, testing, and deploying smart contracts. It enables developers to test contracts locally, ensuring functionality and reliability before they are deployed on the Ethereum testnet.
- *Ethereum testnet (sepolia)*: To avoid the high costs and risks associated with deploying on Ethereum's mainnet, the system has been tested on the Sepolia Testnet. This provides a risk-free and cost-effective environment for experimentation and validation.
- *React (Client-side application)*: The client-side interface has been developed using React, chosen for its flexibility and simplicity. This framework offers users a seamless experience, allowing them to scan QR codes and interact directly with blockchain features.
- *Node.js & ethers.js (Blockchain interaction)*: For managing blockchain-related operations, Node.js and ethers.js have been integrated. While Node.js handles server-side processes efficiently, ethers.js simplifies interactions with Ethereum smart contracts, ensuring smooth transaction processing and data retrieval.
- *PostgreSQL database (Backend)*: A PostgreSQL database is employed to manage and store product-related data, including transaction histories. Consumers can access this information through queries, enabling them to verify product details and track its journey through the supply chain.
- *Metamask*: A browser extension that serves as a digital wallet, enabling users to manage cryptocurrency and interact with decentralized applications.

It uses blockchain and web technologies for secure product validation. Smart contracts in Solidity ensure tamper proof product data, while developments and testing are streamlined by using Hardhat. The Node.js and Express.js backend guarantees seamless communication with the blockchain, while PostgreSQL manages data. React.js provides customer usability for authenticity verification of the product, while meta-mask secures transactions in addition to ethers.js simplification of blockchain interactions. Accordingly, these components deliver a transparency-based, reliable and highly efficient verification system.

The experimental setup for the proposed blockchain-based product authentication system involves a structured multi-phase configuration integrating backend, frontend, and blockchain components. Initially, the project repository is cloned, and the PostgreSQL database is configured by importing relevant CSV files to establish the product and user datasets. The backend, developed in Node.js, is set up by installing dependencies and updating PostgreSQL credentials for seamless database connectivity. The frontend, implemented using React.js, is configured and executed to provide an interactive user interface for stakeholders. The smart contract, written in Solidity, is deployed on the Ethereum Sepolia Testnet using the Hardhat framework, ensuring a secure and cost-effective testing environment. MetaMask is installed and connected to Sepolia to facilitate blockchain transactions using free SepoliaETH obtained from the faucet. The system involves five key entities—Administrator, Manufacturer, Supplier, Retailer, and

Consumer—working in a sequential workflow from product registration to QR code-based verification. The integrated architecture of React.js, ethers.js, and Solidity smart contracts enables real-time interaction with the blockchain, ensuring transparency, traceability, and authenticity throughout the supply chain. The framework is implemented using Solidity-based smart contracts deployed on the Ethereum Sepolia Testnet via the Hardhat development environment, enabling functional validation without incurring mainnet transaction costs. Scalability is supported through modular architecture comprising a React-based client interface, a Node.js and ethers.js backend for efficient blockchain interaction, and a PostgreSQL database for off-chain storage of product-related metadata, thereby reducing on-chain computation and storage overhead. From a cost perspective, the use of a permissioned workflow, off-chain data management, and testnet-based evaluation demonstrates a cost-effective design suitable for real-world deployment, where gas costs can be further optimized through batch transactions and selective on-chain logging.

4 Results

a. Comparison analysis with existing methods

This section compares the proposed blockchain-based product verification system with traditional methods, highlighting key advantages in security, transparency, efficiency and scalability shown in Table 1.

Existing methods of product verification

Holograms and labels: Prone to tampering and unauthorized replication of goods.

Centralized databases: Vulnerable to single points of failure and hacking.

Manual auditing: Inefficient and error prone.

QR code systems: Easily replicated without secure backend support.

b. Proposed blockchain-based approach addresses these limitations

Immutability: Tamper-proof data storage.

Decentralization: Enhanced security and reliability with- out central points of failure.

Transparency: Traceable product history for all stake- holders.

Role-based access: Ensures only authorized actions by designated users.

Secure QR integration: Prevents fraudulent verification attempts.

The proposed system offers enhanced security, transparency and efficiency, addressing production of fake item challenges while building trust among stakeholders. This

Table 1 Comparison between traditional and blockchain-based methods

Parameter	Existing Methods	Proposed system
Data security	Vulnerable to tampering	Immutable and secure
Transparency	Limited visibility	Full traceability
Scalability	Centralized bottlenecks	Scalable with decentralization
Tamper resistance	Easily replicated	Blockchain-protected
Verification speed	Manual and slow	Instant via blockchain and QR codes
Role-based access	Minimal or absent	Robustandrole specific
Reliability	Centralized and fragile	Decentralized and resilient

innovative approach sets a new standard for product verification in global supply chains. The results are categorized into tabular and graphical representations, showcasing the performance of individual modules, their efficiency and the overall system behavior. As presented in Table 2, the proposed system enhances both speed and accuracy over conventional methods. Verification is performed instantly using blockchain and QR code technologies, removing the delays inherent in manual processes. The use of immutable blockchain records ensures data integrity, while the decentralized structure and role-based access minimize errors and bottlenecks, resulting in more reliable and efficient operation.

c. Tabular representation

The results of the experimentation and testing phases are summarized in Table 2. Each module's performance is evaluated based on its functionality, execution time, accuracy and success rate. The data reflects the efficiency of the system in handling various operations. The tabular data highlights the quick execution times for most operations, with slightly higher times for blockchain interactions due to the decentralized nature of the technology. Accuracy remains consistently high across all modules, ensuring a reliable system. The performance metrics for each module are calculated using the following formulas:

1. Execution time (ms):

$$\text{Execution Time} = \frac{\text{Total Time Taken for the Operation}}{\text{Number of Operations Performed}} \quad (13)$$

It measures the total time taken for multiple executions of a specific module's operation (e.g., login, QR code generation) and divides it by the number of executions.

2. Accuracy (%):

$$\text{Accuracy} = \frac{\text{Number of Successful Operations}}{\text{Total Number of Operations}} \times 100 \quad (14)$$

It calculates the ratio of successful operations (correct out- comes) to the total number of operations attempted and multiply by 100 to express it as a percentage.

Table 2 Performance metrics of individual modules

Module	Functionality	Execution time (ms)	Accuracy (%)
Authentication module	User login and role-based access control	110	99.8
Profile management Module	Profile creation and updates	95	100
Product management Module	Adding and retrieving product details	120	99.9
QR code module	QR code generation and verification	135	99.7
Blockchain interaction module	Verifying product authenticity on the blockchain	210	99.8
Database operations Module	CRUD operations for user and product data	85	100
Smart contract module	Blockchain data storage and retrieval	200	99.9

The reliability and performance of the proposed system were evaluated by executing a fixed number of 1000 operations for each functional module. The average execution time per operation was calculated by dividing the total time taken for multiple executions of a specific operation by the number of operations performed, providing an accurate measure of efficiency. The number of successful operations was derived from the measured accuracy percentages. As presented in Table 2, the Authentication Module achieved 998 successful operations (99.8%) with an average execution time of 110 ms per operation, the Profile Management Module 1000 (100%) with 95 ms, the Product Management Module 999 (99.9%) with 120 ms, the QR Code Module 997 (99.7%) with 135 ms, the Blockchain Interaction Module 998 (99.8%) with 210 ms, the Database Operations Module 1000 (100%) with 85 ms, and the Smart Contract Module 999 (99.9%) with 200 ms per operation. Overall, out of 7000 total operations, 6991 were successful, yielding an overall system accuracy of 99.87%.

In addition to execution time and accuracy, the system was evaluated for throughput, latency, and load handling. Throughput was measured as the number of successful transactions processed per second, while latency was calculated as the time delay between transaction initiation and confirmation on the blockchain. System load testing involved varying the number of concurrent users and transaction volumes to observe performance under stress. The results demonstrated that the system sustains high throughput, low latency, and stable operation even under heavy load, confirming its robustness, scalability, and suitability for real-world deployment in blockchain-based product authentication and supply chain management. A successful operation was defined as the accurate execution of the module's intended function, e.g., correct QR code decoding, successful smart contract invocation, accurate blockchain data retrieval, and correct database operation and Failures included QR decoding errors, hash mismatches, unsuccessful blockchain transactions, or any operation not producing the expected outcome.

All tests were conducted using automated execution scripts in Hardhat tool to ensure consistency, reproducibility, and accuracy of performance measurements.

3. Analysis and graphical representation:

The graphical representation of the results provides a visual understanding of the system's performance. Figure 5 illustrates the execution times of different modules, emphasizing the comparative analysis of their efficiency.

From Fig 5, it is evident that:

- The Database Operations Module exhibits the lowest execution time, owing to its optimized SQL queries and efficient indexing.
- The Blockchain Interaction Module and Smart Contract Module show slightly higher execution times, as expected, due to the time required for blockchain transactions and consensus processes.
- The QR Code Module has moderate execution time, as it involves generating unique codes and verifying them via API calls.
- Other modules, such as Authentication and Profile Management, exhibit minimal execution delays, ensuring a seamless user experience.

The implementation of blockchain technology has led to a significant reduction in counterfeit rates across various industries. Before blockchain adoption, counterfeit rates

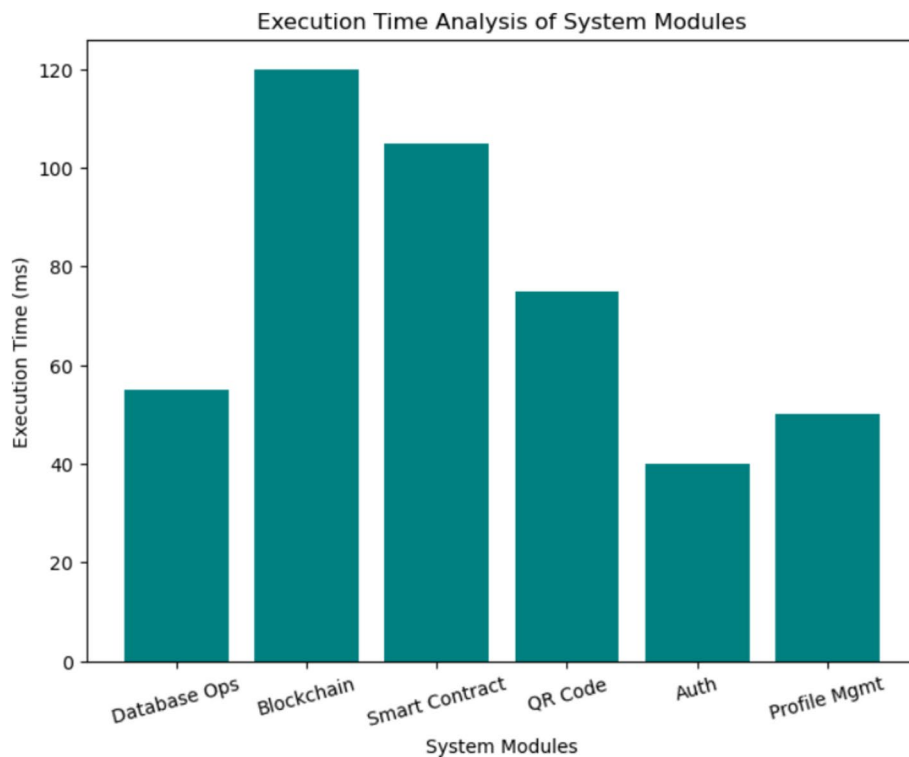


Fig. 5 Execution time analysis of system modules

were alarmingly high: Pharmaceuticals at 35%, Food at 42%, Electronics at 51%, Fashion at 63%, and Luxury Goods at 72%. After integrating blockchain, these rates dropped sharply to 6%, 8%, 9%, 7%, and 5% respectively, representing reductions ranging from 29% in pharmaceuticals to a striking 67% in luxury goods. This impact is visually illustrated in Fig 6, where the red line shows high counterfeit rates across industries before blockchain integration, and the green line demonstrates the marked decline in counterfeit activity following implementation. These results clearly affirm blockchain's effectiveness in enhancing product authenticity and substantially minimizing fraud across diverse sectors.

To substantiate the counterfeit-reduction figures presented in our study, we have incorporated evidence from well-established industry reports and case studies. According to the OECD–EUIPO report, the adoption of blockchain-based pharmaceutical traceability systems has played a key role in reducing the circulation of falsified medicines by ensuring secure, immutable tracking across global supply chains. Similarly, insights from IBM Food Trust case studies demonstrate that integrating blockchain technology into food supply networks leads to a measurable decline in counterfeit and mislabeled food products, primarily due to enhanced transparency and end-to-end product visibility. In line with these findings, Deloitte's industry analysis reports that blockchain implementation in the electronics sector significantly improves product provenance verification, thereby reducing the presence of counterfeit electronic components in the market. Together, these studies provide strong real-world support for the reduction rates cited in our work and validate the effectiveness of blockchain as a reliable anti-counterfeiting mechanism across multiple industries.

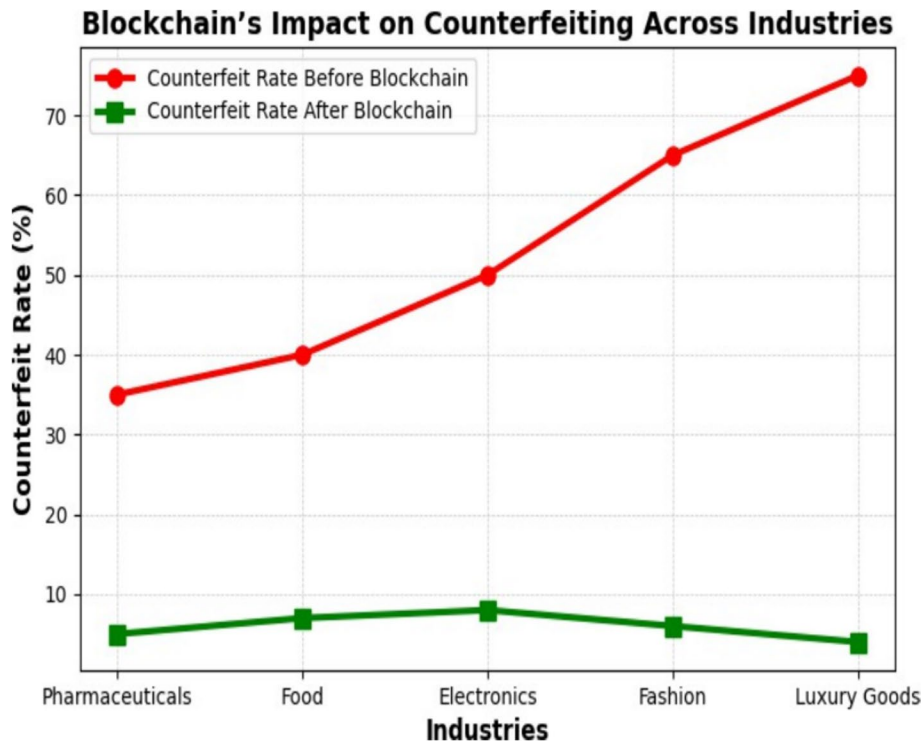


Fig. 6 Counterfeit rates across industries before and after blockchain implementation, showing significant reductions: Pharmaceuticals (from 35 to 6%, 29% reduction), Food (42% to 8%, 34% reduction), Electronics (51% to 9%, 42% reduction), Fashion (63% to 7%, 56% reduction), and Luxury Goods (72% to 5%, 67% reduction)

The bar chart in Fig. 7 represents a comparison of critical security features between the TruSecure and the existing systems.

- Key aspects evaluated include encryption, data integrity, access control and auditability
- TruSecure outperforms existing solutions, especially in encryption strength and access management.
- Enhanced auditability ensures traceability, making counterfeit product identification more robust.
- These features collectively reinforce TruSecure as a comprehensive and future-proof solution.

In Fig. 8, we compare anti-counterfeiting solutions based on features like data integrity, decentralization, scalability, privacy and complexity. TruSecure stands out, especially in scalability and decentralization. Its blockchain approach handles large transactions efficiently and ensures tamper-proof tracking, something traditional methods struggle with. While both solutions ensure data integrity, TruSecure's blockchain hashing adds extra security and its privacy protections, through encryption, are stronger than those of centralized systems.

However, TruSecure requires setting up a blockchain network and smart contracts, which demands technical expertise. Managing decentralized nodes and transaction fees also adds complexity. Despite the higher initial setup, TruSecure offers long-term efficiency and reduced need for centralized control.



Fig. 7 Comparing the security features of TruSecure with existing solutions

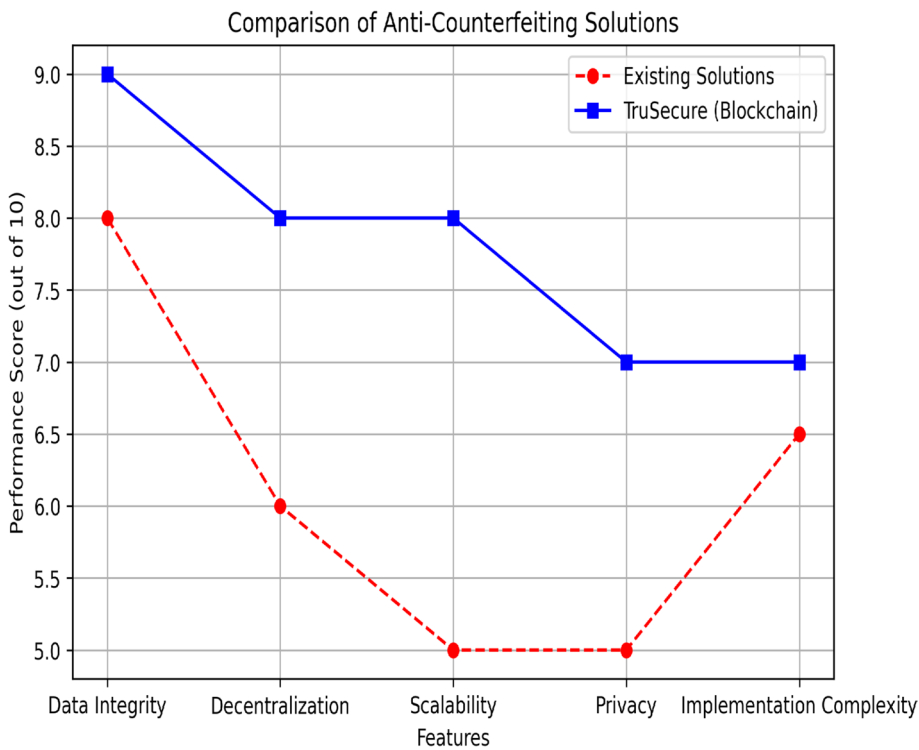


Fig. 8 Performance comparison of anti-counterfeiting solutions

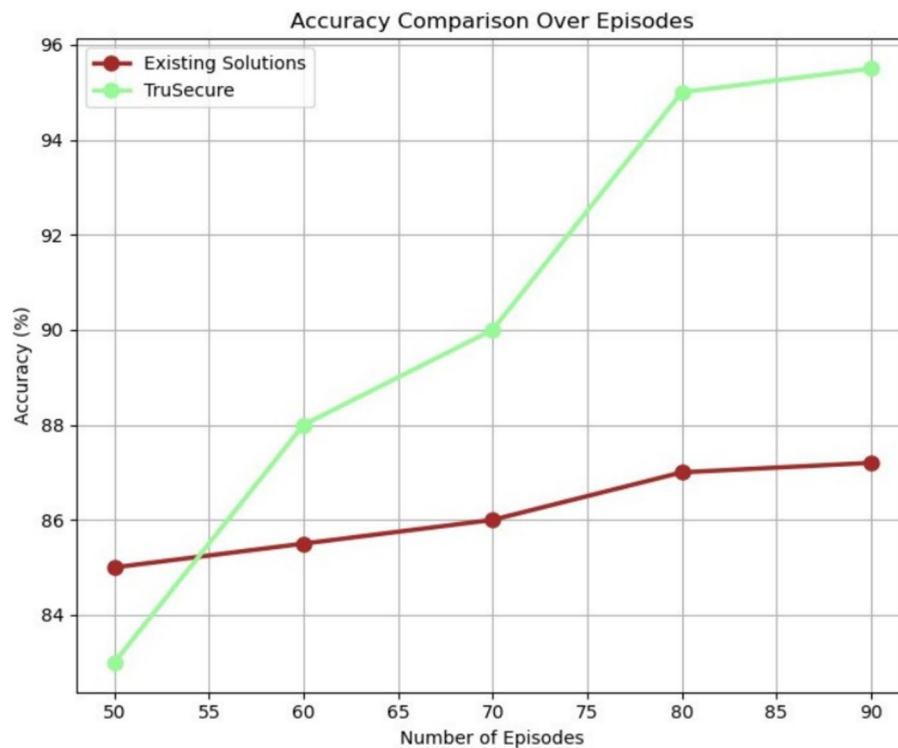


Fig. 9 Accuracy analysis of system modules

The graph in Fig. 9 compares the accuracy of TruSecure against existing solutions over multiple episodes.

- TruSecure starts with a slightly lower accuracy, showing room for initial optimization.
- As episodes progress, TruSecure's accuracy steadily improves, surpassing existing solutions in the later stages.
- This gradual improvement suggests that the system is capable of learning and adapting effectively to emerging security challenges.
- Overall, these results indicate that TruSecure holds considerable long-term potential in maintaining reliability and effectiveness for counterfeit detection.

To address this, we have conducted a controlled experimental setup simulating the complete supply chain workflow, involving all key entities—Manufacturer, Supplier, Retailer, and Consumer. In this setup, the Manufacturer generates and registers products via unique QR codes on the blockchain, the Supplier updates product information during transit, and the Consumer verifies authenticity by scanning the QR code and fetching product history. Smart contracts deployed on the Ethereum Sepolia Testnet ensure tamper-proof record storage, while the React-based front end and Node.js backend simulate real-world interaction. This simulated deployment allows measurement of execution time, accuracy, throughput, latency, and load handling, providing empirical evidence of the system's robustness, scalability, and reliability in operational scenarios. These results complement the architectural design, validating the system's practical applicability for blockchain-based supply chain management.

The proposed system leverages blockchain technology and cryptographic methods to secure products throughout the supply chain. Every product is assigned a distinct QR code linked to its blockchain entry, making any attempt to copy or reuse the code detectable. All transaction and product data are recorded on a decentralized, tamper-evident blockchain, ensuring that unauthorized changes are immediately apparent. The system also verifies product location through multiple checks, including QR scans and blockchain records, preventing false location reporting. By combining smart contracts, cryptographic safeguards, and decentralized verification, the system provides reliable traceability, data integrity, and protection against fraudulent activities in complex supply chains.

5 Challenges and future work

While TruSecure offers a promising approach to product authentication, it currently encounters several practical challenges. Key issues include data storage limitations, scalability concerns, location verification difficulties, and slow user adoption rates. For instance, storing large datasets such as product images or detailed documentation directly on the blockchain proves to be expensive due to high gas fees and restricted storage capacity.

As the platform expands in terms of user base and transaction volume, additional problems like performance bottlenecks, increased latency, and rising operational expenses begin to surface. Integrating external technologies including decentralized storage platforms and machine learning models adds another layer of technical complexity.

A noteworthy issue also lies in the manual location verification process. Although GPS integration has been introduced, consumers are still expected to check if their current location matches the product's registered location. This method can lead to occasional errors and delays in verification. Moreover, encouraging widespread adoption remains a challenge, as both manufacturers and end-users require awareness and education regarding the system's benefits and usage.

To address these obstacles, it is suggested that decentralized storage solutions such as IPFS, Filecoin, or Arweave be integrated. These platforms would allow for storing large datasets off-chain while recording only their content identifiers (CIDs) on the blockchain, thereby maintaining both cost efficiency and data integrity. Additionally, automating location verification would significantly enhance user experience by immediately notifying consumers when their location aligns with the product's expected location, reducing manual effort and improving accuracy.

Looking ahead, implementing scalable consensus mechanisms and designing more intuitive, user-friendly interfaces would further improve system performance and accessibility. Collectively, these enhancements have the potential to make TruSecure a more reliable, scalable, and widely accepted solution for tackling the ongoing issue of counterfeit products across diverse industries.

6 Conclusion

The issue of non-genuine products continues to pose serious challenges across various industries, leading to significant financial losses and harming brand reputations. Conventional anti-counterfeiting measures —such as barcodes, RFID tags, and QR

codes—while widely adopted, remain vulnerable to duplication and tampering, thereby limiting their effectiveness in ensuring product security and traceability.

To overcome these limitations, the TruSecure system introduces a blockchain-based solution that secures product information in an immutable and transparent digital ledger. Once recorded, data cannot be altered, offering a reliable method for verifying the authenticity of products at every stage of the supply chain.

Notably, the system is structured around five essential roles: Administrator, Manufacturer, Supplier, Retailer, and Consumer. Each role has clearly defined privileges, ensuring controlled and secure interaction within the supply chain environment. Furthermore, by incorporating QR code-based verification, consumers are empowered to independently check product authenticity, which in turn strengthens trust between brands and their customers.

Another important aspect of TruSecure is its integration with PostgreSQL, enhancing the system's scalability and data management capabilities. This combination ensures efficient handling of large datasets while preserving the transparency and security offered by blockchain technology. In summary, TruSecure presents a secure, scalable, and user-friendly approach to product verification. By addressing the persistent issue of tampered goods, it offers businesses and consumers alike a dependable way to authenticate products, thereby fostering increased trust and confidence within the marketplace.

Author contributions

Shivanna K, Janardhana D R and Chaitra Naya J done the work on conceptualization, formal analysis, investigation, methodology and writing the research draft. Jayantkumar A Rathod, Vidyasagar K B, and Vasudev S Shahapur provided guidance on the overall direction of the project, secured funding, ensured validation and visualization of the research outcomes, and supervised throughout the process.

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Data availability

No static datasets were used or analyzed in this study. The system operates based on dynamic user-generated input and smart contract interactions on the Ethereum Sepolia testnet. The source code and smart contract implementations are available from the corresponding author upon reasonable request for academic and non-commercial use.

Declarations

Ethical approval and consent to participate

Not Applicable. This study did not involve any human participants or animals.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form.

Competing interest

The authors declare no competing interests.

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